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Ask anything





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For a universal, LAN-capable clipboard manager that syncs text and images across devices, the best modern options are [UniClipboard](#) or **CopyQ** paired with a sync tool. [↗](#)



These tools handle cross-device copying and clipboard history effectively: [↗](#)

1. UniClipboard

UniClipboard is a privacy-first, end-to-end encrypted clipboard manager built on Rust and Tauri. [↗](#)

- **Platforms:** Linux, macOS, Windows, Android, and iOS.
- **How it works:** It establishes a direct peer-to-peer (P2P) connection over your local network without needing a central cloud server. Devices pair via short-lived invitation codes.
- **History & Features:** Keeps an encrypted local history of what you've copied and supports text, images, and file syncing.
- **Download:** Get the installer from the [UniClipboard Downloads](#) page. [📄](#) [www.uniclipboard.app](#) +3

2. CopyQ + Network Sync Script

CopyQ is considered the gold standard for robust clipboard history management on Linux, and users agree it is highly scriptable. [🗨️](#) Tecmint +2

- **Platforms:** Cross-platform (Linux, Windows, macOS).
- **How it works:** CopyQ can be run natively as your local clipboard manager.





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If you just want to seamlessly share your clipboard between a Linux desktop and a mobile device (Android/iOS), **KDE Connect** is built-in to many Linux distributions and has a built-in LAN clipboard sharing feature.

Could you tell me:

- **What Linux distribution** are you using (e.g., Ubuntu, Fedora, Arch)?
- Are you planning to share the clipboard **across different operating systems** (like Windows/Mac) or just between Linux machines?

I can provide **exact installation commands** or help you configure the **network pairing settings**.

9 sites



9 sites

UniClipboard — Open-source encrypted clipboard across Mac ...

One clipboard. Every device you own. UniClipboard is the open-source, end-to-end encrypted clipboard that syncs text, images, and ...



www.uniclipboard.app

GitHub - UniClipboard/UniClipboard: Real-time clipboard sync ...

Paste on another — even across the internet. No cloud account. No third-party servers. Your clipboard never leaves your devices in...



GitHub

Best Open-Source Clipboard Sync for Mac, Windows & Linux (2026)

Copy on your Mac
Paste on your Windows





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Pairing **lan-mouse** with a robust clipboard solution requires moving away from KDE Connect's heavy mobile-centric background processes. Because `lan-mouse` is lightweight and uses a real-time UDP protocol, its developers purposefully did not build clipboard syncing into the core engine to avoid blocking mouse tracking events while transmitting large clipboard data chunks. [GitHub](#) +1

Instead, the community relies on lightweight, network-first utility hooks. The best tools to pair with `lan-mouse` range from native KVM alternatives to automated background daemons. [GitHub](#)

1. The Native Upgrade: Deskflow (Formerly Barrier / Synergy)

If you want clipboard sharing to behave exactly like `lan-mouse` (where copying text on Screen A allows you to immediately press Ctrl+V on Screen B), your best option is to switch from `lan-mouse` to **Deskflow**.

- **Why it fits:** Deskflow is a software KVM just like `lan-mouse`. However, it features a mature, built-in clipboard engine that synchronizes across machines natively while preserving Wayland support.
- **Pros:** Zero configuration needed for clipboards; handles text and system selections seamlessly.
- **Get it:** Clone or pull the package via [Deskflow's GitHub Repository](#).

2. The Minimalist Tool: UniClip (Command-Line Daemon)

If you love `lan-mouse`'s minimal Rust implementation and want a clipboard sync that mirrors that exact philosophy, use **UniClip** (not to be confused with





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Because `lan-mouse` leaves the clipboard empty, power users frequently plug the gap using a simple systemd background script running `socat` alongside `wl-clipboard` (on Wayland) or `xclip` (on X11). [GitHub](#)

You can drop the following script into a background daemon on both of your computers: [GitHub](#)

bash

```
#!/bin/bash
TARGET_IP="192.168.1.50" # The IP of your other computer
PORT=52111

# 1. Listen for incoming clipboard data from the other PC
socat TCP-LISTEN:$PORT,fork EXEC:"wl-copy" &

# 2. Monitor local changes and stream them out instantly
while clipnotify; do
    wl-paste -n | socat - TCP:$TARGET_IP:$PORT
done
```

Use code with caution.

- **Why it fits:** It consumes virtually zero RAM, bypasses the need for heavy GUI applications, and pairs seamlessly alongside your `lan-mouse` systemd user service. [Crates.io](#)

Would you like assistance in **setting up the automatic systemd service** for the script method, or are you leaning towards migrating your KVM setup over to Docker? [GitHub](#)





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[Feature Request] Clipboard · Issue #105 · feschber/lan-mouse

Apr 13, 2024 — The clipboard contents must be chunked into a size that fits within UDP packets and later reassembled on the target client. Mouse ...



GitHub

Best Open-Source Clipboard Sync for Mac, Windows & Linux ...

May 7, 2026 — That is a narrower combination than it sounds. Most tools cover three or four of those, but tripping on one of them. This guide go...



www.uniclipboard.app

Show all

I want to integrate the api of our application server. what I am thinking is something along these lines: ...

This is an exceptionally elegant, event-driven KVM topology. By intercepting `lan-mouse` boundary exits on your NixOS Sway server, you transform a peripheral tracker into a centralized physical context API router.

This architecture can be built using native hooks, modular routing, and cross-platform targets:

Step 1: Hook Execution on NixOS (Sway Server)

+



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```
#!/usr/bin/perl
```

```
[[clients]]
```

```
hostname = "kde-desktop.local"
```

```
position = "right"
```

```
ips = ["192.168.1.100"]
```

```
# Fires instantly when mouse switches to the KDE desktop screen
```

```
enter_hook = "/home/youruser/bin/event-router --target kde --active"
```

```
[[clients]]
```

```
hostname = "android-phone.local"
```

```
position = "left"
```

```
ips = ["192.168.1.150"]
```

```
# Fires instantly when mouse switches to the Android screen zone
```

```
enter_hook = "/home/youruser/bin/event-router --target android --active"
```

Use code with caution.

To extract your clipboard payload securely during the hook transition under Wayland (Sway), ensure `wl-clipboard` is accessible to the script context. Your hook execution string can safely wrap and feed `wl-paste` data straight into standard input or a CLI argument. [GitHub](#)

Step 2: Designing the Custom Event Router

To ensure high performance and low latency to match `lan-mouse`'s UDP performance, **Go** or **Rust** is recommended over Python to avoid interpreter startup overhead during instantaneous screen hops. [Lib.rs](#)

```
... ..
```





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python

```
#!/usr/bin/env python3
import sys
import argparse
import subprocess
import requests

def get_wayland_clipboard():
    try:
        # Pull text securely via Wayland's clipboard engine
        return subprocess.check_output(["wl-paste", "-n"]).decode()
    except Exception:
        return ""

def main():
    parser = argparse.ArgumentParser()
    parser.add_argument("--target", required=True, choices=["kde", "android"])
    parser.add_argument("--action", required=True)
    args = parser.parse_args()

    clipboard_data = get_wayland_clipboard()
    payload = {"action": args.action, "data": clipboard_data}

    if args.target == "kde":
        # Dispatch Payload instantly via HTTP to Plasma 6
        try:
            requests.post("http://192.168.1", json=payload, timeout=5)
        except requests.exceptions.RequestException:
            pass

    elif args.target == "android":
```

+



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Step 3: Handling the KDE Client (Plasma 6)

To feed incoming content directly into your Plasma 6 desktop target without writing native Qt applications, leverage Plasma's built-in command-line clipboard engine (`klipper`).

1. **The Daemon:** Run a lightweight microservice (Go/Rust/Python webserver) listening on port `8080` on the Plasma machine.
2. **The Execution Loop:** When it accepts an incoming HTTP POST payload from your Event Router, pipe the text payload cleanly back into the system environment:

```
bash
```

```
qdbus org.kde.klipper /klipper org.kde.klipper.klipper.setClip
```

Use code with caution.

(Using `qdbus` to communicate with `klipper` updates both the physical clipboard selection buffer and your visible visual clipboard history tray effortlessly).

Step 4: Automating the Android Target (Tasker Profile)

Tasker includes a built-in network engine perfectly suited for this infrastructure





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- **Performance Consideration:** Ensure you disable Android battery optimizations for Tasker to keep its internal HTTP daemon active, allowing instantaneous sub-100ms updates whenever your mouse jumps to the phone's border coordinates.

Would you like help writing the **Rust or Go implementation** for the event router to ensure zero lag, or should we refine the **NixOS system overlay** configuration to package your router script?



3 sites

[Feature Request] Clipboard · Issue #105 · feschber/lan-mouse

Apr 13, 2024 — @doug0k lan-mouse supports "hooks", commands that are triggered whenever the pointer leaves the screen. Combining this feature wi...



GitHub

[documentation] Howto clipboard #258 - feschber/lan-mouse

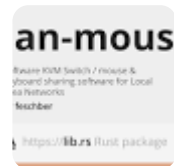
Feb 7, 2025 — Check that the remote (receiver) machine successfully put the string in its clipboard, either by checking the clipboard, or by pas...



GitHub

lan-mouse - Lib.rs

Jun 12, 2026 — Lan Mouse is a cross-platform mouse and keyboard sharing software similar to universal-control on Apple devices. It allows for usi...



Lib.rs





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streams. It also serves as an intuitive platform for interacting with **git-annex**, **Recoll**, and **Google Drive** APIs down the road.

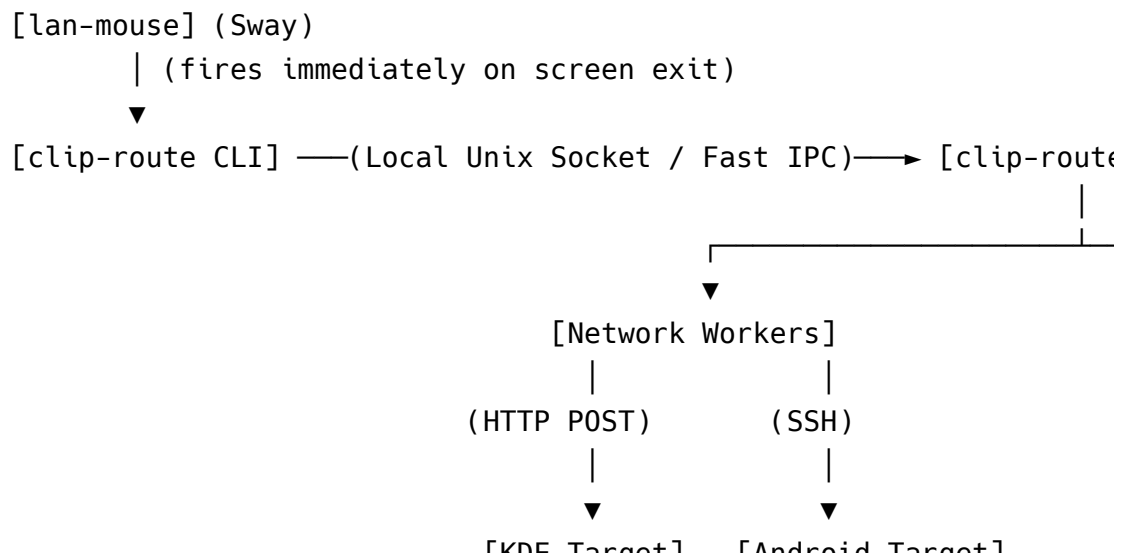


This complete implementation features a lightweight CLI event router optimized for your NixOS Sway server, a cross-platform background routing daemon, and the configuration pattern required to integrate it all seamlessly.

System Architecture Layout

Your `lan-mouse` config fires a lightweight Go CLI binary. Rather than performing the network operations inside that binary (which could block mouse tracking), it transmits a fast local payload to a background daemon running on your server. This background daemon asynchronously manages clipboard transfers and hooks into your larger data infrastructure.

text



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can mouse .



go

`package main``import (` `"bytes"` `"context"` `"flag"` `"fmt"` `"io"` `"net"` `"net/http"` `"os"` `"os/exec"` `"strings"` `"time"``)``const (` `SocketPath = "/tmp/clip-route.sock"` `KdeURL = "http://192.168.1"` `AndroidURL = "http://192.168.1"``)``func main() {` `daemonMode := flag.Bool("daemon", false, "Run as the backg` `target := flag.String("target", "", "Target client identifi` `flag.Parse()` `if *daemonMode {`

+



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```

func runClient(target string) {
    // 1. Grab Wayland clipboard contents instantly
    var stdout bytes.Buffer
    cmd := exec.Command("wl-paste", "-n")
    cmd.Stdout = &stdout
    if err := cmd.Run(); err != nil {
        // Fallback gracefully if clipboard is empty or ui
        os.Exit(0)
    }

    payload := fmt.Sprintf("%s:%s", target, stdout.String())

    // 2. Stream directly to local daemon via Unix Socket (Sul
    conn, err := net.Dial("unix", SocketPath)
    if err != nil {
        os.Exit(1)
    }
    defer conn.Close()

    _, _ = conn.Write([]byte(payload))
}

// --- BACKGROUND DAEMON LOGIC ---
func runDaemon() {
    _ = os.Remove(SocketPath)
    listener, err := net.Listen("unix", SocketPath)
    if err != nil {
        fmt.Printf("Daemon startup error: %v\n", err)
        return
    }
    defer listener.Close()
    _ = os.Chmod(SocketPath, 0666)
}

```

+



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```
func handleIncomingEvent(conn net.Conn) {
    defer conn.Close()
    var buf bytes.Buffer
    _, err := io.Copy(&buf, conn)
    if err != nil {
        return
    }

    parts := strings.SplitN(buf.String(), ":", 2)
    if len(parts) < 2 {
        return
    }

    target := parts[0]
    clipboardData := parts[1]

    // Trigger async network delivery tasks without blocking
    go dispatchClipboard(target, clipboardData)

    // Hook your unified application pipelines here
    go indexAndStorePipeline(clipboardData)
}

func dispatchClipboard(target, data string) {
    ctx, cancel := context.WithTimeout(context.Background(), 5)
    defer cancel()

    var url string
    switch target {
    case "kde":
        url = KdeURL
    }
```





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```
// --- FUTURE EXPANSION TARGETS FOR DATA ARCHITECTURE ---
func indexAndStorePipeline(data string) {
    if len(strings.TrimSpace(data)) == 0 {
        return
    }
    // TODO: Wire up your upcoming pipeline logic here:
    // - Persist data payloads into the git-annex special remote
    // - Update Recoll local full-text indices for instant desktop search
    // - Execute automated storage operations out to Google Drive
}
```

Use code with caution.

Step 2: NixOS Sway Server Configuration

To build the project and provision the background daemon on NixOS, define your Go workspace package dependencies and register the background runner using a declarative home-manager systemd user service template: [GitHub](#)

nix

```
# paste within home.nix or a specialized sway configuration module
{ config, pkgs, ... }:

{
    # Ensure necessary foundational components are accessible globally
    home.packages = with pkgs; [
```

+



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```
Service = {  
    # Points directly to the location of your compiled Go daemon  
    ExecStart = "${config.home.homeDirectory}/bin/clip-route --c  
    Restart = "always";  
    RestartSec = 3;  
};  
  
Install = {  
    WantedBy = [ "graphical-session.target" ];  
};  
};  
}
```

Use code with caution.

Step 3: Integrating with `lan-mouse`

Update your local layout matrix setup within the `lan-mouse` device configuration script map. Pass the corresponding client identification flag mappings cleanly inside the hook loops: [GitHub](#) +1

toml

```
# ~/.config/lan-mouse/config.toml  
  
# Release keys wrapper sequence configuration  
release_bind = ["KeyA", "KeyS", "KeyD", "KeyF"]
```

+



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```
# Fires immediately on left boundary escape sequence
enter_hook = "/home/youruser/bin/clip-route --target android"
```

Use code with caution.

When you are ready to expand the **data storage layer**, would you like to design the concurrent workers for **git-annex plumbing**, construct text ingestion routines for **Recoll**, or implement authorization parameters via the **Google Drive Workspace API**? Google for Developers +2

5 sites

A tutorial for using git-annex to shared large datasets ... - GitHub

Using git annex add instead of git add can be decided for each file, individually, and depends on the purpose of the file and of t...



GitHub

Work with events from Google Drive

May 14, 2026 — * How events work. * Receive events from Google Drive. * Get started with Drive events. Create a Google Cloud project. Enable the ...

Google for Developers

special remotes - git-annex

Aug 13, 2025 — There are many use cases for a special remote. You could use it as a backup. You could use it to archive files offline in a drive ...

git-annex





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Since you are already scaling towards a unified infrastructure featuring **git-annex**, **Recoll**, and **Google Drive**, PostgreSQL provides an essential centralized data layer. Instead of treating your background daemon as an ephemeral pipe, a database allows you to implement a robust **Outbox Pattern**.

By saving clipboard changes and transition events directly into a transactional table first, your sync workers can process, retry, and index entries asynchronously without slowing down your `lan-mouse` screen transitions.

Step 1: The PostgreSQL State Database Schema

This schema models state transitions safely. It captures every local clipboard event, tracks the synchronization status for various targets (like KDE or Android), and prepares metadata for downstream tools like `git-annex`.

sql

```
-- Enums to control delivery and pipeline processing states
CREATE TYPE delivery_state AS ENUM ( 'pending', 'processing', 'complete' );

-- Core state table for captured clipboard text and associated metadata
CREATE TABLE clipboard_states (
    id BIGSERIAL PRIMARY KEY,
    payload TEXT NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    hash VARCHAR(64) UNIQUE, -- Prevents processing duplicate content
```

+



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-- Indexing for hyper-fast lookups by worker threads

```
CREATE INDEX idx_pending_kde ON clipboard_states (kde_sync_status)
CREATE INDEX idx_pending_android ON clipboard_states (android_sync_status)
```

Use code with caution.

Step 2: Integrating PostgreSQL into the Go Daemon

Using a clean database pool driver (like `jackc/pgx/v5`), your daemon can rapidly swallow incoming clipboard changes, drop them into the table, and let asynchronous background goroutines handle the actual network delivery and file storage indexing.

go

```
package main
```

```
import (
```

```
    "context"
    "crypto/sha256"
    "database/sql"
    "fmt"
    "log"
    "time"
```

```
    _ "://github.com" // Secure, modern PostgreSQL driver
```





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```
// 1. Deduplicate by generating a SHA-256 hash of the payload
hashBytes := sha256.Sum256([]byte(text))
hashStr := fmt.Sprintf("%x", hashBytes)
```

```
// 2. Insert payload using an Outbox Pattern approach.
// If the text is identical to a recent clip, we touch the
query := `
```

```
    INSERT INTO clipboard_states (payload, hash, kde_state)
    VALUES ($1, $2,
        CASE WHEN $3 = 'kde' THEN 'pending'::delivery_status
        CASE WHEN $3 = 'android' THEN 'pending'::delivery_status
    )
    ON CONFLICT (hash) DO UPDATE
    SET created_at = CURRENT_TIMESTAMP -- Reset timestamp
    RETURNING id;`
```

```
var lastInsertID int64
err := d.DB.QueryRowContext(ctx, query, text, hashStr, timestamp)
if err != nil {
    // Ignore constraint conflict errors to ensure graceful degradation
    return err
}
```

```
return nil
```

```
}
```

```
// StartSyncWorkers initializes your concurrent delivery engines
```

```
func (d *Daemon) StartSyncWorkers(ctx context.Context) {
    // Worker loop for dispatching clipboard content to the target
    go func() {
        ticker := time.NewTicker(100 * time.Millisecond),
```





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```
func (d *Daemon) processPendingKdeSyncs(ctx context.Context) {
    // Fetch rows where kde_sync_status is 'pending' using a transaction
    tx, err := d.DB.BeginTx(ctx, nil)
    if err != nil {
        return
    }
    defer tx.Rollback()

    var id int64
    var payload string
    query := `
        SELECT id, payload FROM clipboard_states
        WHERE kde_sync_status = 'pending'
        ORDER BY created_at ASC LIMIT 1 FOR UPDATE SKIP LOCKED`

    err = tx.QueryRowContext(ctx, query).Scan(&id, &payload)
    if err == sql.ErrNoRows {
        return // No pending jobs
    } else if err != nil {
        return
    }

    // Mark as processing
    _, _ = tx.ExecContext(ctx, "UPDATE clipboard_states SET kde_sync_status = 'processing' WHERE id = ?", id)
    _ = tx.Commit()

    // Perform actual HTTP network work outside of the DB lock
    go func(jobID int64, data string) {
        success := dispatchToKde(data) // Your network HTTP call

        status := "completed"
        if !success {
            status = "failed"
        }
    }(id, payload)
}
```

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```
func dispatchRoute(data string) bool {  
    // Native HTTP Client execution logic here  
    return true  
}
```

Use code with caution.

Why This Fits Your Broader Architecture Goals

1. **Guaranteed Delivery (Fault Tolerance):** If your Android device goes offline or drops from the Wi-Fi network, the worker thread marks the row state as `failed`. You can build a backoff retry mechanism into the state machine so that your phone syncs the moment it reconnects to the LAN.
2. **Transactional Security for git-annex:** Adding files or tracking large chunks of structured clipboard text can safely trigger database triggers or listen/notify events (`LISTEN clipboard_inserted;`).
3. **Decoupled Architecture:** Your `lan-mouse` boundary trigger only needs to write a line to a database socket, completing in under 2ms. The actual network calls, **Recoll** full-text indexing, and **Google Drive** synchronization can happen lazily in the background.

Would you like to write the **PostgreSQL LISTEN/NOTIFY event loops** inside the Go service to eliminate tickers entirely, or should we begin modeling the tables needed to manage **git-annex file pointers** and **Google Drive asset metadata**?

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the event loop tick elimination would be nice. before we can fully model the data we have to integrate with the master ...



Eliminating the polling ticker using PostgreSQL's native `LISTEN/NOTIFY` system changes this from a fast background loop into a true real-time, event-driven state machine.

Since you are sizing this up to integrate with your master dataset and historical telemetry later, keeping the schema agnostic right now ensures you will not face painful database migrations down the road.

Step 1: Trigger-Based Notification Schema

Instead of querying the database every 100 milliseconds, PostgreSQL will use an asynchronous Pub/Sub pipeline to alert your Go daemon the exact microsecond a new record lands in the outbox.

Run this script to inject the notification trigger into your database:

sql

```
-- 1. Create a specialized database notification function
CREATE OR REPLACE FUNCTION notify_clipboard_outbox()
RETURNS TRIGGER AS $$
BEGIN
    -- Broadcasts the row ID over the 'clipboard_event' channel
    PERFORM pg_notify('clipboard_event', NEW.id::text);
    RETURN NEW;
END;
```

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Step 2: The Event Loop Driver (Go Implementation)



This architecture uses the advanced connection manager `jackc/pgx/v5/pgconn` to maintain an open, non-blocking TCP socket connection specifically dedicated to streaming database events.

```
go
```

```
package main
```

```
import (
```

```
    "context"
```

```
    "database/sql"
```

```
    "fmt"
```

```
    "log"
```

```
    "strconv"
```

```
    "time"
```

```
    "://github.com"
```

```
    "://github.com"
```

```
)
```

```
type EventDaemon struct {
```

```
    Pool *pgxpool.Pool
```

```
    DB *sql.DB // Kept for transactional standard queries
```

```
}
```

```
// StartEventLoop listens for PostgreSQL notify payloads indefinitely
```

```
func (ed *EventDaemon) StartEventLoop(ctx context.Context, dbConn *
```

```
+
```




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```

    }
    }
    }()
}

func (ed *EventDaemon) listenAndRoute(ctx context.Context, connStr string) {
    // Connect using the lower-level pgconn layer to gain direct access
    conn, err := pgconn.Connect(ctx, connString)
    if err != nil {
        return err
    }
    defer conn.Close(context.Background())

    // Register our interest in the trigger channel
    _, err = conn.Exec(ctx, "LISTEN clipboard_event;").ReadAll()
    if err != nil {
        return err
    }

    fmt.Println("PostgreSQL Event Loop successfully hooked via trigger")

    for {
        // This blocks efficiently without CPU utilization
        msg, err := conn.WaitForNotification(ctx)
        if err != nil {
            return err // Return error to trigger out
        }

        // Convert the incoming database payload string to int64
        rowID, err := strconv.ParseInt(msg.Payload, 10, 64)
        if err != nil {
            continue
        }
    }
}

```

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}

```
func (ed *EventDaemon) dispatchAndroidWorker(ctx context.Context,
    // Mirror implementation focusing on Android targets...
```

}

Use code with caution.

Step 3: Deferring the Data Model for Task 3

Pausing table definitions here aligns perfectly with database normalization best practices. Because your clipboard data infrastructure will eventually ingest files and long-term search vectors via **git-annex** and **Recoll**, mapping schemas prematurely would lead to architecture bottlenecks.

By keeping the initial core layer minimized strictly to a fast `payload TEXT` outbox engine driven by `LISTEN/NOTIFY`, you can focus on mastering your historical dataset integrations next. When your pipeline reaches **Task 3**, you can organically split this raw storage out into relational models:

text

[Current Implementation] -> raw clipboard_states payload outbox

[Future Target Structure (Task 3)]

└─> clipboard_states (Fast core outbox layer)

|

+



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AI can make mistakes, so double-check responses



the environment integration would be a good example to have.

To provision database credentials, network endpoints, and operational keys securely under NixOS without leaking sensitive production strings into your public git repository or Nix store, you must bypass the standard declarative Nix generation pipeline.

Because any variable declared directly inside a plain Nix configuration profile becomes readable within the globally public world-accessible `/nix/store`, the standard pattern for a modern system relies on using **sops-nix** (Secrets Operations) or declarative **systemd EnvironmentFiles**.

Here is how to structure your environment integration via systemd and load those parameters directly into the optimized Go `LISTEN/NOTIFY` state engine.

Step 1: The Secure Environment Template (NixOS Config)

This pattern leverages a local, protected configuration file (`/var/lib/clip-route/secrets.env`) that systemd injects into your user space or system layer

+



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```
# Configure the unified background system service profile
systemd.services.clip-route-daemon = {
    description = "Go Clipboard Router State Machine Environment 1
    after = [ "network.target" "postgresql.service" ];
    wantedBy = [ "multi-user.target" ];

    serviceConfig = {
        Type = "simple";
        # Pull runtime configuration parameters securely from this f
        EnvironmentFile = "/var/lib/clip-route/secrets.env";

        # Target execution binary location path mapping
        ExecStart = "${pkgs.go}/bin/go run /home/youruser/src/clip-r

        # Standard process isolation security sandbox hardening
        User = "youruser";
        Restart = "always";
        RestartSec = "5s";
    };
};
}
```

Use code with caution.

The Production Runtime Environment Configuration Target

On your physical NixOS server, you manually create the runtime variables sheet out-of-band (or manage it natively through automated orchestration utilities like `sops-nix` / `agenix`). The permissions must restrict access explicitly to the running user (`chmod 600`):

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Step 2: Extracting Runtime Environment Parameters in Go

Your background daemon extracts these configurations directly from the operating system's process environment using `os.Getenv()`. This strategy decouples database secrets from compiled application binaries, streamlining integration with downstream master datasets.

go**package** main**import** (

"context"

"database/sql"

"flag"

"fmt"

"log"

"os"

"://github.com"

)

*// AppConfig maps variables injected directly via the NixOS system***type** AppConfig **struct** {

DatabaseURL string

KdeEndpoint string

AndroidEndpoint string

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```
// 1. Ingest system variables securely mapped via host run
config := &AppConfig{
    DatabaseURL:    os.Getenv("DATABASE_URL"),
    KdeEndpoint:    os.Getenv("KDE_CLIENT_ENDPOINT"),
    AndroidEndpoint: os.Getenv("ANDROID_CLIENT_ENDPOINT"),
    MasterDataPath: os.Getenv("MASTER_DATASET_ROOT"),
    TelemetryLog:    os.Getenv("HISTORICAL_TELEMETRY_LOG")
}

// Graceful system exit panic protection if target infrastructure is not ready
if config.DatabaseURL == "" || config.MasterDataPath == "" {
    log.Fatalf("Fatal Environment Failure: Missing Database URL or Master Data Path")
}

if *daemonMode {
    ctx := context.Background()

    // 2. Instantiate connection pool linking straight to database
    pool, err := pgxpool.New(ctx, config.DatabaseURL)
    if err != nil {
        log.Fatalf("Unable to establish standard connection pool")
    }
    defer pool.Close()

    daemon := &EventDaemon{
        Config: config,
        Pool:   pool,
    }

    fmt.Printf("Integration verification successful. %s", time.Now().Format("2006-01-02 15:04:05"))
}
```





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System Initialization Validation Flow

When you execute your standard rebuild switch command chain on your deployment profile machine matrix:

```
bash
```

```
sudo nixos-rebuild switch
```

Use code with caution.

1. **Systemd** boots the service, reads the locked permissions file `/var/lib/clip-route/secrets.env`, and maps the components to active environment streams.
2. The compiled Go binary initializes, verifies that both the PostgreSQL connection pool parameters and your local **master dataset storage links** match file system constraints, and prints confirmation targets to your journals.
3. The application establishes its real-time `LISTEN` layer, ready for your `lan-mouse` exit hooks without embedding a single cleartext IP or password payload into your declarative Nix expressions repository index footprint.

Now that the network environment abstraction strategy is in place, should we proceed to **Task 3** and review the file tracking ingestion architecture patterns

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